第十五章量子物理单元测试题参考答案

一、选择题(共 30 分)

1.A. 光电效应的光子能量不能小于逸出功 eU_0，$h\nu \geq eU_0$，$h\frac{c}{\lambda} \geq eU_0$

2.D. 遏止电压 $eU_a = \frac{1}{2}m\upsilon_{\max}^2 = h\nu - A$，$\Delta U_a = \frac{1}{e}h\Delta\nu = \frac{hc}{e}(\frac{1}{\lambda_2} - \frac{1}{\lambda_1})$

3.D. 单色光频率不变，则 $\frac{1}{2}m\upsilon_{\max}^2 = h\nu - A$ 不变；电势差不变，则 $\Delta\frac{1}{2}m\upsilon^2 = eU$ 不变

4.B. $\Delta\lambda = \lambda - \lambda_0 = \frac{h}{m_0 c}(1-\cos\varphi)$

5.D. 能量守恒 $h\nu_0 + m_0c^2 = h\nu + mc^2$，电子动能 $E_K = mc^2 - m_0c^2 = h\nu_0 - h\nu$

$$\frac{E_K}{\varepsilon} = \frac{h\nu_0}{h\nu} - 1 = 0.2$$

6.D. $$\frac{mc^2 - m_0c^2}{m_0c^2} = \frac{1}{\sqrt{1-\frac{\upsilon^2}{c^2}}} - 1 = \frac{1}{\sqrt{1-0.6^2}} - 1$$

7.C. $m\upsilon r = n\hbar$，$\frac{\upsilon_1}{\upsilon_3} = \frac{n_1 r_3}{n_3 r_1}$，而 $r_n = n^2 r_1$ $\frac{\upsilon_1}{\upsilon_3} = \frac{n_3}{n_1}$

8.A. $\Delta E = E_n - E_1$，$E_n = 10.9 - 13.6 = -3.41(eV)$ $h\nu = E - E_n = -0.85 + 3.41$，

9.A. $\lambda = \frac{h}{p}$

10.C. $p = \frac{h}{\lambda}, \Delta p = \frac{h}{\lambda^2}\Delta\lambda, \Delta p_x \Delta x = \frac{h}{\lambda^2}\Delta\lambda\Delta x \geq h, \Delta x \geq \frac{\lambda^2}{\Delta\lambda}$

二、填空题(共 21 分)

1. 不变 ； 变长 ； 波长变长 .

2. 1.8

$$\frac{1}{\lambda} = R_H(\frac{1}{2^2} - \frac{1}{n^2}), \frac{1}{\lambda_{\max}} = R_H(\frac{1}{2^2} - \frac{1}{3^2}), \frac{1}{\lambda_{\min}} = R_H(\frac{1}{2^2} - \frac{1}{\infty^2})$$

3. 1.51

$$\Delta E = 0 - \frac{-13.6}{n^2} = \frac{13.6}{9}$$

4. 3.29×10^{-21} J

$p = \frac{h}{\lambda}$， $E_k = \frac{p^2}{2m} = \frac{h^2}{2m\lambda^2}$

5. 1.33×10^{-23}

$$\Delta p_x \geq \frac{h}{\Delta x}$$

三、计算题(共 49 分)

1. 解：设能使该金属产生光电效应的单色光最大波长为λ_0.

由 $h\nu_0 - A = 0$

可得 $(hc/\lambda_0) - A = 0$

$\lambda_0 = hc/A$

又按题意： $(hc/\lambda) - A = E_K$

∴ $A = (hc/\lambda) - E_K$

得 $$\lambda_0 = \frac{hc}{(hc/\lambda) - E_K} = \frac{hc\lambda}{hc - E_K\lambda} = 612 \text{ nm}$$

2. 解：入射光子的能量为 $$\varepsilon_0 = \frac{hc}{\lambda_0}$$

散射光子的能量为 $$\varepsilon = \frac{hc}{\lambda}$$

反冲电子的动能为 $$E_K = \varepsilon_0 - \varepsilon = hc(\frac{1}{\lambda_0} - \frac{1}{\lambda}) = 1.68\times10^{-16}$$

3. 解：设始态能级量子数为 k, 则轨道半径由 r_k 变为 r_n, 且 $r_k = qr_n$.

由 $$r_k = k^2 \frac{\varepsilon_0 h^2}{\pi m e^2}$$

可得 $$k^2 = qn^2$$

光子的频率 $$\nu = Rc(\frac{1}{n^2} - \frac{1}{k^2})$$

即 $$\nu = \frac{Rc}{n^2}(1 - \frac{n^2}{k^2}) = \frac{Rc}{n^2}(1 - \frac{1}{q})$$

4. 解：所发射的光子能量为 $\varepsilon = hc/\lambda = 2.56$ eV

氢原子在激发能为 10.19 eV 的能级时，其能量为

$$E_K = E_1 + \Delta E = -3.41 \text{ eV}$$

氢原子在初始状态的能量为 $E_n = \varepsilon + E_K = -0.85$ eV

该初始状态的主量子数为 $$n = \sqrt{\frac{E_1}{E_n}} = 4$$

5. 解：非相对论动能 $$E_K = \frac{1}{2} m_e \upsilon^2$$

而 $p = m_e\upsilon$ 故有 $$E_K = \frac{p^2}{2m_e}$$

又根据德布罗意关系有 $p = h/\lambda$ 代入上式

则 $$E_K = \frac{1}{2} h^2/(m_e\lambda^2) = 4.98\times10^{-6} \text{ eV}$$